

INTRODUCTION

CH 1

General Anatomy

- Anatomy: which deals with Structure of human body.

* Subdivisions of Anatomy:

- (a) Regional. (4 v) (b) Systems.
- (1) Cadaveric Anatomy.
 - (2) Living Anatomy:
 - (i) inspection.
 - (ii) palpation.
 - (iii) percussion.
 - (iv) auscultation.
 - (v) Endoscopy.
 - (3) Embryology.
 - (4) Histology.
 - (5) Surface Anatomy.
 - (6) Radiography and imaging Anat (CT Scan, XRay, MRI).
 - (7) Comparative Anatomy.
 - (8) Physical Anthropology.
 - (9) Applied Anatomy.
 - (10) Experimental Anatomy.
 - (11) Genetics.

* Anatomical Terminology.

- Anatomical position.
- Supine position.
- prone position.
- Lithotomy position.

* planes: (1) median, Sagittal.
mid Sagittal.

↳ divides body Right & left.

(2) Frontal, coronal plane.

↳ divides body Anterior & Posterior.

(3) Transverse, Horizontal → Divides body upper and lower.
Plane

- (8) Invag
- (9) par
- (12) Flex
- (13) Exten
- (14) prox
- (15) dis
- (16) * Ch
- (2) * Ang
- * Gl
- * sp
- pr
- p
- I
- ▲ J
- (1) F
- (2) E
- (3)
- (4)

(4) Oblique plane: → divides body other than "3" planes.
(5) Cardinal plane: → passes to the center of the body.

* **positions & directions:**

- (1) ventral or Anterior: Towards front. [lies around Toe or hallux]
- (2) Dorsal or posterior: Towards back.
- (3) medial: Towards medial line. (10) [medial border]
- (4) lateral: Away from medial line. (11) [lateral border]
- (5) proximal / Cranial / Superior: Close to the head.
- (6) Distal / caudal / Inferior: lower end of trunk. [close to feet]
- (7) Superficial: Close to skin.
- (8) Deep: away from the skin.
- (9) Invagination: Inside projection. (12) [lateral border lies around the little Toe]
- (10) palmar: In front of palm. (13) → Evagination.
- (11) Flexor: Front of upper limb / Back of lower limb.
- (12) Extensor: Back of upper limb / Front of lower limb.
- (13) proximal in lower limb: Close to the root of limb (L₁, L₂)
- (14) Distal in lower limb: Away from the root of limb (L₁, L₂)

* **Other Terms:** (1) Cranial or Rostral: Towards head.
(2) Caudal: Towards Tail.

* **Angular Movements:** • In flexion Angle decreases. (Angle of Ligaments/Tendons)
• In Extension angle Increases.

* **Gliding Movement:**

Circumduction

Includes: • Adduction.
• Abduction.
• Flexion.
• Extension.

* **special Movements:**

- pronation and Supination at radioulnar joint.
- protrusion and Retraction of temporomandibular joint.
- Inversion and Eversion of Subtalar joint.
- ▲ In upper limb: **Shoulder joint** (Possible movements).

- | | |
|----------------------|--------------------|
| (1) Flexion. | (5) Adduction. |
| (2) Extension | (6) Abduction. |
| (3) medial rotation. | (7) Circumduction. |
| (4) lateral rotation | |

- **Elbow Joint:** ① Flexion.
② Extension.
③ Supination. *occurs*
④ pronation. *Elbow Int*

- **Wrist Joint:** ① Flexion.
② Extension.
③ Adduction/Radial deviation.
④ Abduction/Ulnar deviation.

- **THUMB:** ① Opposition of Thumb. *Movements of Fingers*
② Circumduction.
③ flexion.
④ Extension.
⑤ Abduction.
⑥ Adduction.
- ① Adduction → when all fingers get together.
② Abduction → when all fingers
③ Flexion → when we make fist.
④ Extension → when we open fist.

Lower limb.

• **HIP Joint:**

- ① Flexion of HJ
- ② Extension of HJ
- ③ Adduction of HJ
- ④ Abduction of HJ
- ⑤ Circumduction.
- ⑥ medial Rotation.
- ⑦ lateral Rotation.

• **Knee Joint.**

- ① Flexion of knee.
 - ② Extension of knee.
- **Movement of Foot:**
① Dorsiflexion.
② Plantar flexion.
③ Inversion.
④ Eversion.

• **TRUNK:** ① Flexion

- ② Extension
- ③ lateral flexion.
- ④ lateral rotation.

- **Neck:** ① Flexion: when Face comes closer to Chest.
② Extension: when Face Taken Away from the Chest.
③ lateral flexion: when Ear close to Shoulder.
④ Rotation.

- **Mouth** ① opening of mouth.
② closure of mouth.
③ protraction.
④ Retraction.

ANATOMY

cutting up.
- To study the structure of body part and their relation is called Anatomy.

↓
GROSS ANAT ↓ microscopic Anat ↓ Developmental Anat ↓ specialized branches of Anat

① Gross Anatomy

i) Regional ⇒ All structure in one frame.

ii) Systemic ⇒ gross anatomy of the body studied by system.

iii) Surface ⇒ Study of internal structure as relate to the over line.

② MICROSCOPIC ANATOMY

i) Cytology ⇒ Study of cells.

ii) Histology ⇒ Study of Tissues.

③ Developmental Anatomy

i) Embryology: Study of developmental changes in birth.

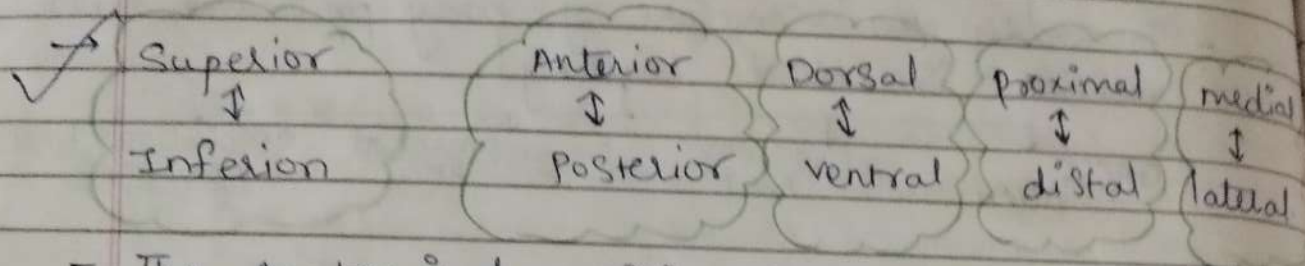
④ Specialized Branches of Anatomy

i) pathological Anatomy: Study of structural changes cause by disease.

ii) Radiographic Anatomy: visualised by X-Ray.

iii) Study of Anatomical Structure at a Sub-cellular level.

★ Some Anatomical Directions.



- The Anatomical position is a standardised method of observing or imaging the body that allows precise and consistent anatomical reference, when in the Anatomical position the subject stands.
- Standing up right → Facing the observer, head level, eyes facing forward, feet flat on floor, arms at the side, palms forward (ventral).

★ position and directions:-

- Terms of position and direction describes the position of two parts relative to one another of one body.
- It usually along one of the 3 measure body plane.
- **Superior**: refers to a structure being closer to the head OR higher than another structure in the body.
- **Inferior** refers to a structure being closer to the feet or lower than another structure in the body.
- **Anterior**: refers to structure being more in front than another structure in the body.
- **Posterior** refers to the structure being more in back than another structure in the body.

- **Medial:** refers to the structure closer to the midline or medial plane of the body than another structure.
- **Lateral:** refers to the structure being far away from the midline than another structure.
- **Dorsal:** Towards the Back (like dorsal fin of Dolphin)
- **Ventral:** Towards the front OR Belly.
- **Distal:** Refers to the structure being further away from Root of the limbs than another structure in the limb.
- **Proximal:** Refers to the structure being closer to the root of limbs than another structure in that limb.
- **Superficial:** refers to the structure being closer to the surface body than another structure.
- **Deep:** refers to a structure being closer to the core of the body than another structure.
- **Supine:** lying face up.
- **Prone:** lying face down.
- **Unilateral:** pertaining to one side of the body.
- **Bilateral:** pertaining to both sides of the midline body.

Sleeping positions only.

* Basic Body Planes OR Sections.

These terms are used for planes or section that cut the body, organs, tissues or cells.

- 1) Transverse plane
 - 2) frontal plane
 - 3) midsagittal plane
- Anatomical Planes.

- Fixed line to reference along which the body is often divided or sectioned to facilitate viewing of its structures.
- Allow one to obtain a three dimensional perspective by studying the body from different views.

① Sagittal plane: The plane dividing the body into right and left portion.

- mid Sagittal or median are names for the plane dividing the body into equal right and left halves

② Frontal plane: The plane which divides the body into front and Back portions.

- also called as coronal plane
- divide the body into anterior & posterior section.

③ Transverse plane: The horizontal plane which divides body into upper & lower portion.

- Also called as horizontal plane
- Divide the body into superior/inferior sections